



Uncovering Hidden Patterns in Occupational Injuries

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Introduction

Why this matters

- Millions of workplace injuries annually
- Current analysis is limited with what models were used and what variables were analyzed
- Complex relationships between event type, injury nature, body part, and source get missed

	ID	UPA	EventDate	\	Part of Body Title	Event	\
0	2015010015	931176	1/1/2015		Lower leg(s)	1214	
1	2015010016	930267	1/1/2015		Leg(s), n.e.c.	317	
2	2015010018	929823	1/1/2015		Nonclassifiable	4331	
3	2015010019	929711	1/1/2015		Leg(s), unspecified	640	
4	2015010020	929642	1/1/2015		Finger(s), fingernail(s), n.e.c.	6411	

What our data looks like

- Dataset: OSHA Severe Injury Reports (2015-2025)
- **103,750** records, 28 columns
- Feature categories we used:
 - What happened (Nature of Injury, Event Type)
 - Where it happened (Body Part, Industry NAICS)
 - Why it happened (Hospitalization, Amputation, Loss of eye)

Cleaning

- What we dropped: Address2, Secondary Source, Secondary Source Title, Inspection
- What we filled: Missing numbers → 0, Missing text → "Unknown"
- What we standardized: Binary columns

Missing values percentage:

Address2	91.170984
Inspection	66.274460
Secondary Source	53.808710
Secondary Source Title	53.808710
Longitude	0.065098
Latitude	0.065098
Zip	0.023319
Address1	0.017489
City	0.016517
Amputation	0.006801
Loss of Eye	0.004858
Primary NAICS	0.001943

	Primary NAICS	Hospitalized	Amputation	Loss of Eye	FederalState	NatureTitle	Part of Body Title	combined_text	EventTitle	SourceTitle
0	922140	1	0	0	1	Fractures	Lower leg(s)	Three correctional facility guards were escort...	Injured by physical contact with person while ...	Co-worker
1	339999	1	0	0	1	Second degree heat (thermal) burns	Leg(s), n.e.c.	Employee in the Machine Shop received second d...	Ignition of vapors, gases, or liquids	Welding, cutting, and blow torches
2	484121	1	0	0	1	Traumatic injuries and disorders, unspecified	Nonclassifiable	A truck driver fell approximately 4 feet while...	Other fall to lower level less than 6 feet	Semi, tractor-trailer, tanker truck
3	424490	1	0	0	1	Soreness, pain, hurt-nonspecified injury	Leg(s), unspecified	An employee's leg was pinned between a truck a...	Caught in or compressed by equipment or object...	Pallet jack-powered
4	326122	1	0	0	1	Fractures	Finger(s), fingernail(s), n.e.c.	An employee working on the Line 6 Auto-Beller ...	Caught in running equipment or machinery durin...	Metal, woodworking, and special material machi...

Pipeline

Pipeline:

Raw Data → Preprocessing → Embeddings → VAE → Latent Space (z) → Clustering

Input:

- Categorical → embeddings
- Binary → numeric

Architecture:

- Encoder → latent distribution (μ, σ)
- Latent vector z
- Decoder → reconstructs input

Why VAE?

- Captures nonlinear relationships
- Produces structured latent space
- Improves clustering quality

```
model = OSHAEmbeddingVAE(  
    vocab_sizes=vocab_sizes,  
    embedding_dims=embedding_dims,  
    num_binary=X_bin.shape[1],  
    hidden_dims=HIDDEN_DIMS,  
    latent_dim=LATENT_DIM  
)  
model.to(DEVICE)
```

Results

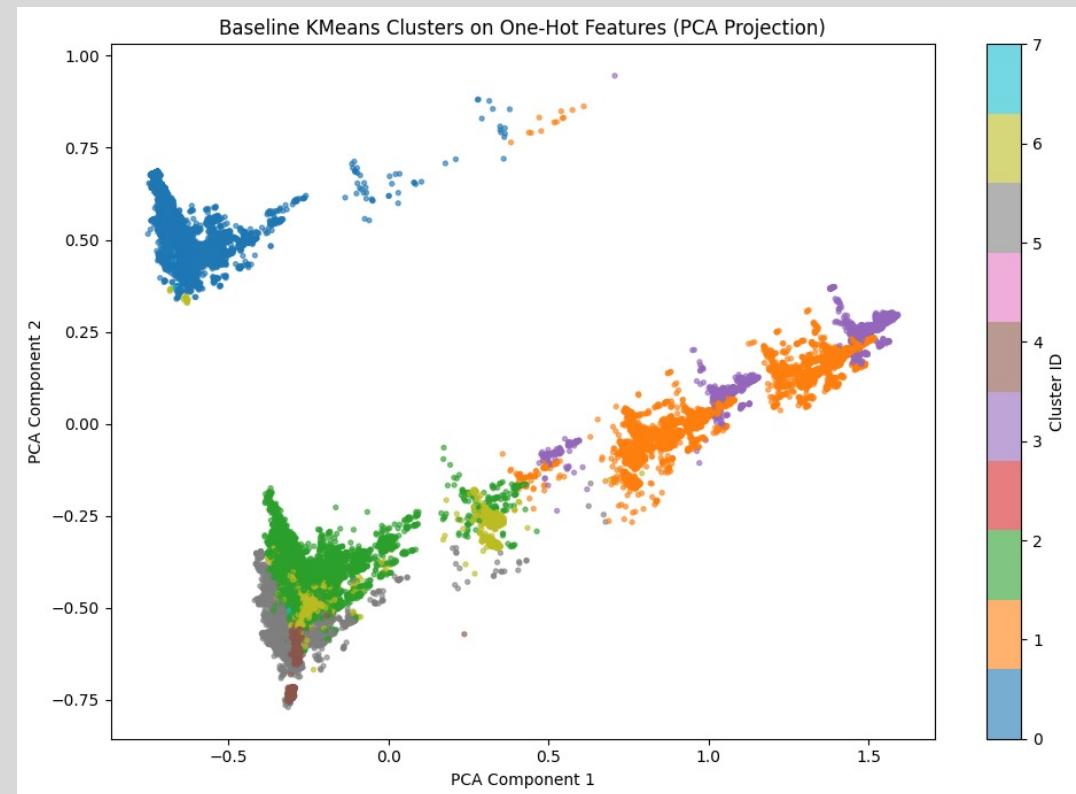
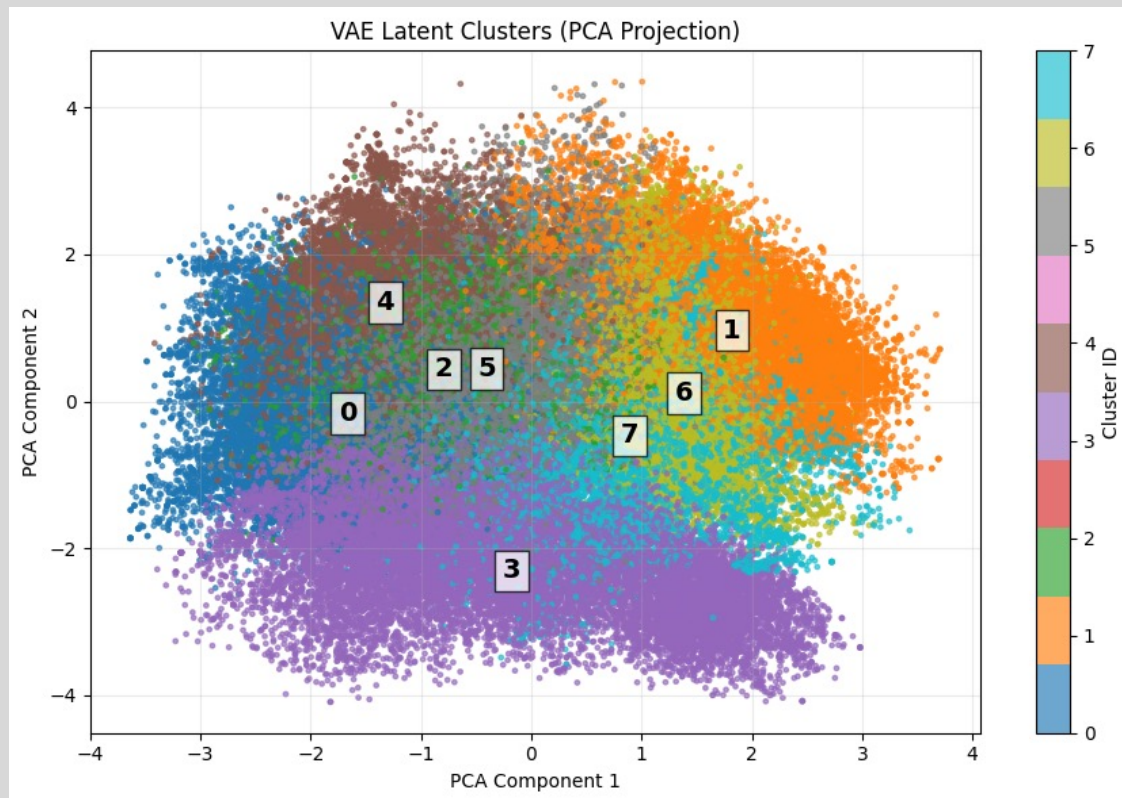
Cluster ID	VAE Size	VAE %	K-Means Size	K-Means %
0	14871	14.33%	32969	31.78%
1	14256	13.74%	16190	15.61%
2	7133	6.87%	25252	24.34%
3	13983	13.47%	10205	9.84%
4	11439	11.02%	4873	4.70%
5	16513	15.91%	6508	6.27%
6	12148	11.71%	6287	6.06%
7	13407	12.92%	1466	1.41%
Total	103750	100%	103750	100%



Cluster ID	VAE Cluster (Semantic Pattern)	Key Characteristics (VAE)	Closest K-Means Cluster	K-Means Interpretation	Key Difference
0	Falls from height (ladders, roofs)	Fractures + ladder/roof sources + multi-height falls	Cluster 0	Fractures + falls	K-Means captures injury type but misses source (ladder/roof)
1	Machinery-related amputations	Finger injuries + machinery + cutting/sawing	Cluster 1 / 3	Amputations + machinery	K-Means splits same scenario into multiple clusters
2	Transport / vehicle injuries	Forklifts + pedestrian accidents + lower body injuries	No clear match	Mixed (falls, heat, objects)	K-Means fails to isolate vehicle-related risks
3	Mixed severe injuries (falls + machinery)	Fractures + amputations + slips + equipment	Cluster 6	Mixed injuries	K-Means cluster is noisy and less interpretable
4	Slip/trip falls (same level)	Hip injuries + indoor floors + slipping/tripping	Cluster 5	Pain + falls	K-Means captures symptom (pain), not cause (slip/trip)
5	Object-related injuries	Struck by objects + compression + equipment	Cluster 2 / 6	Cuts / crushing injuries	K-Means groups by injury, not event mechanism
6	Severe crushing / amputation (equipment)	High amputation rate + compression events	Cluster 3	Amputations	K-Means captures label but misses mechanism (compression)
7	Heat / electrical exposure	Burns + heat + electricity + environmental exposure	Cluster 4	Heat / exposure	One of the few well-aligned clusters

RESULTS

Plots

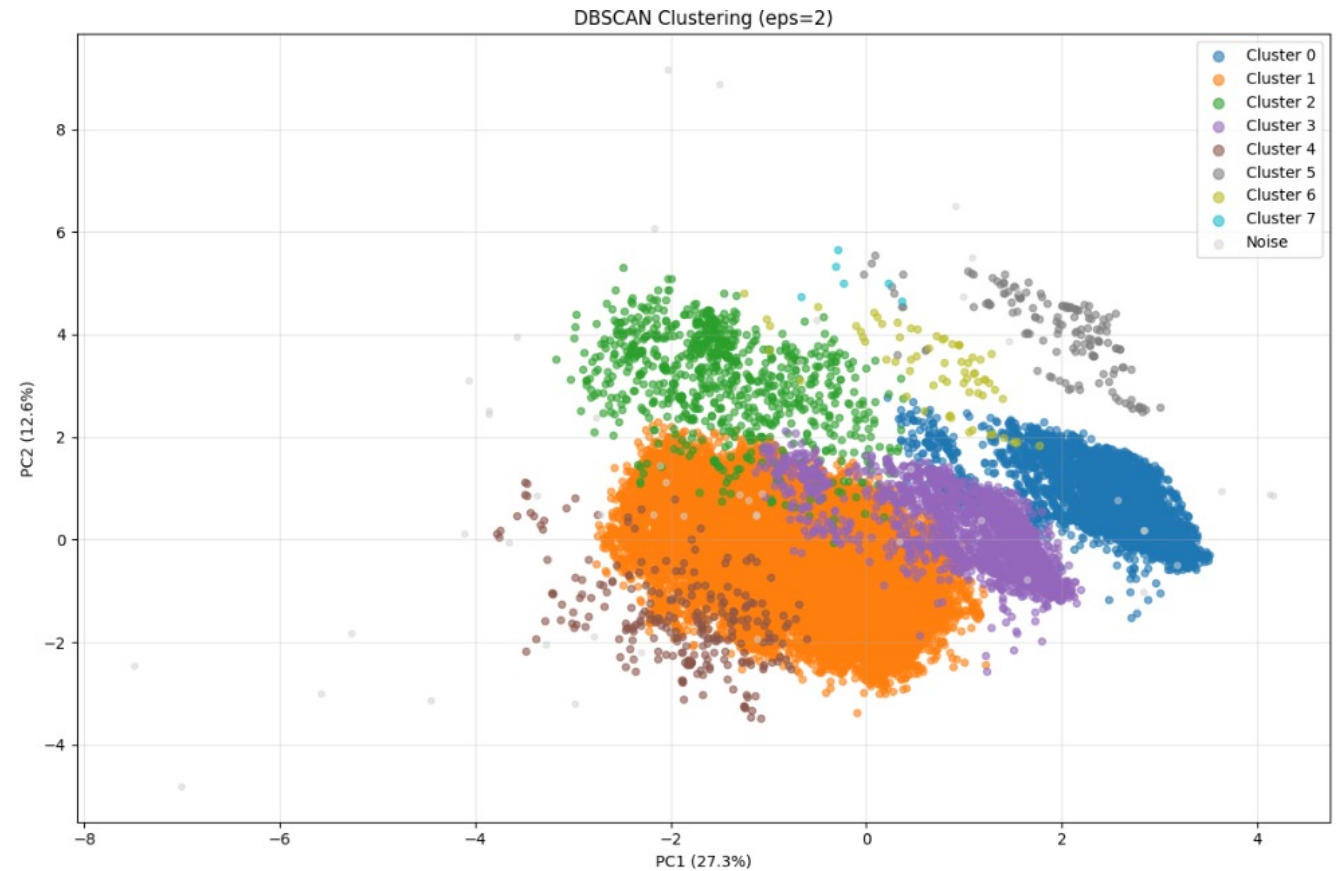


DBSCAN Results

Cluster	Size	Event	Nature	Body Part	Hosp%
1	35734	Other fall to lower level, unspecif	Fractures	Multiple body parts,	100%
0	9260	Caught in running equipment or mach	Amputations	Fingertip(s)	0%
3	3555	Caught in running equipment or mach	Amputations	Finger(s), fingernai	100%
2	925	Exposure to environmental heat	Fractures	BODY SYSTEMS	100%
4	258	Inhalation of harmful substance-sin	Fractures	Nonclassifiable	200%
5	145	Compressed or pinched by shifting o	Amputations	Fingertip(s)	0%
6	66	Compressed or pinched by shifting o	Amputations	Finger(s), fingernai	100%
7	6	Struck by dislodged flying object,	Avulsions, enucleations	Eye(s)	100%

DBSCAN

- One massive main cluster: Most Injuries follow a common pattern
- Separate Clusters: Distinct injury types exist
- Minimal noise (0.1%): Data has strong natural structure
- Small scattered clusters: Rare but identifiable injury patterns



Bonus: What predicts Hospitalizations?

Random Forest (RF) Analysis

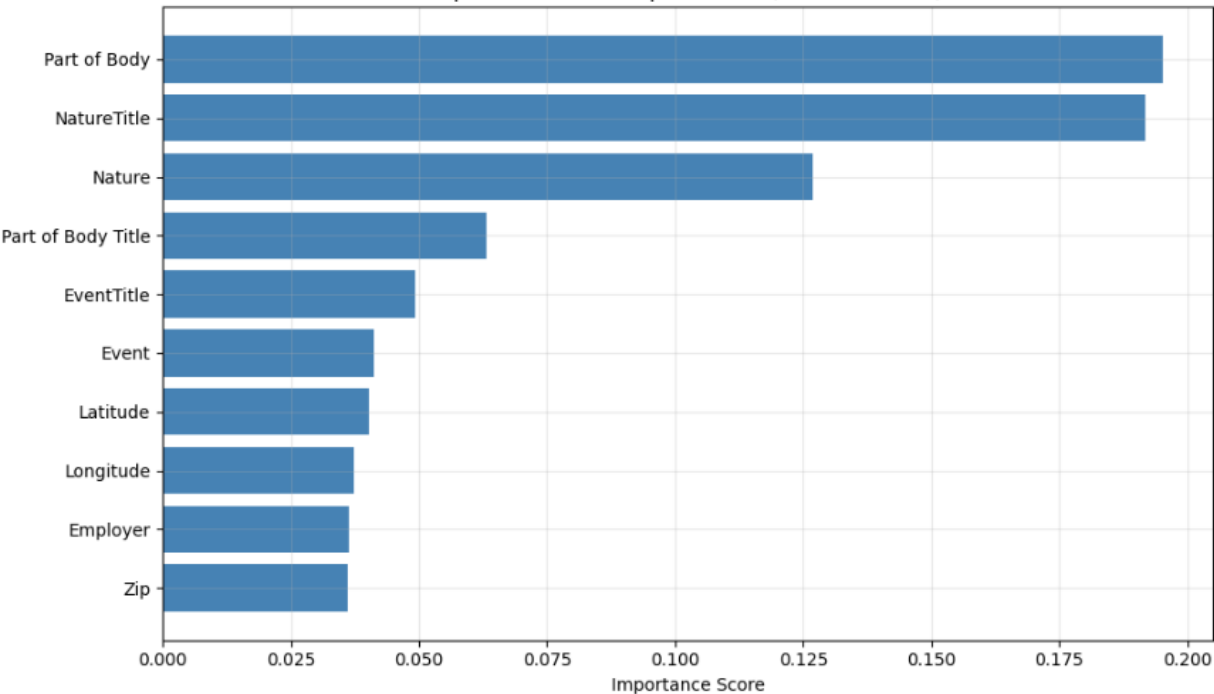
Why Random Forest?

- Validates our previous findings with predictive power
- Identifies which features actually drive severity
- Helps identify actionable insights

Fold accuracy: 0.924
Fold accuracy: 0.922
Fold accuracy: 0.918
Fold accuracy: 0.921
Fold accuracy: 0.919

	Feature	Importance
10	Part of Body	0.195127
9	NatureTitle	0.191691
8	Nature	0.126892
11	Part of Body Title	0.063105
13	EventTitle	0.049301
12	Event	0.041218
5	Latitude	0.040238
6	Longitude	0.037350
0	Employer	0.036325
4	Zip	0.036177

Top Predictors of Hospitalization (Random Forest)



Code & Repository

GitHub Repo: https://github.com/jeevana-swaroop/CS572_Project

GitHub Pages: <https://jeevana-swaroop.github.io/osha-vae-clustering-blog/>

Repository Contents:

- Data preprocessing scripts
- VAE model definition
- Training code
- Clustering and visualization code
- Results and figures

Conclusion

Key Findings

- VAE clustering captures **real-world injury scenarios** by combining event, source, and injury type
- K-Means primarily groups data based on **dominant labels** (e.g., fractures, amputations)
- VAE produces **balanced and diverse clusters**, while K-Means shows **skewed distributions**
- Several meaningful patterns (e.g., **vehicle-related injuries**) are **missed by K-Means**

Core Insight

- VAE learns a **structured latent space**, enabling discovery of **multi-dimensional patterns**, whereas K-Means relies on **surface-level feature similarity**

Practical Impact

- Enables **targeted safety interventions** (e.g., ladder safety, machinery handling, forklift operations)
- Improves **risk understanding** by identifying *causes*, not just outcomes

Final Takeaway

- **VAE-based clustering provides more meaningful, interpretable, and actionable insights compared to traditional K-Means clustering**



THANK YOU!